



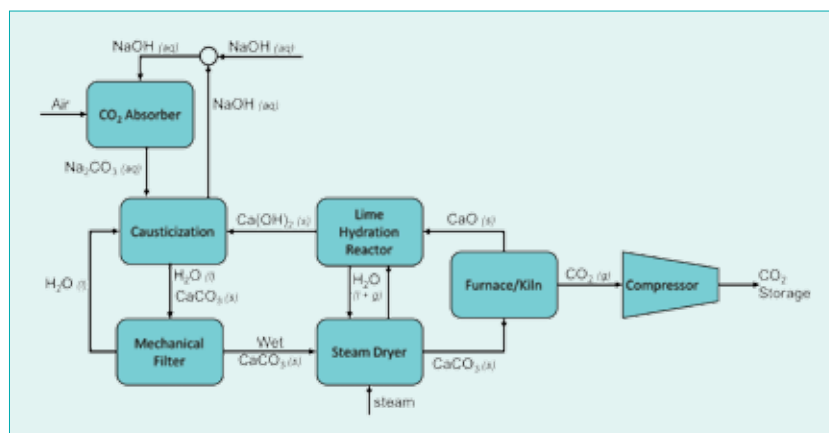
Researchers develop curious AI

Researchers from UC Berkely are teaching AI to be curious, a trait that allows humans to acquire new knowledge. <http://degit.in/curiousai>



Vaping still causes cancer

If you thought vaping lowers the risk of cancer, think again since a new study suggests that it actually increases the risk of heart disease as well. <http://dgit.in/Vapingcncr>



Direct Air Capture Process Flow Diagram using NaOH as the absorbent and including solvent regeneration

sunlight away from Earth, in some pretty creative ways!

Aerosols in the stratosphere

Out of all the methods to reflect sunlight away from the planet, the widespread insertion of aerosols into the upper atmosphere has received the most amount of attention. Models based on this method have shown the ability to offset heating caused by a potential doubling of CO₂ levels (which is the standard used for measuring the effectiveness of such strategies). Due to the pre-existing examples of volcanic eruptions, sulphates are the prime candidates for this process. Risks in this method include changes in the precipitation cycle on the planet and even the depletion of the ozone layer.

Marine cloud brightening

Taking reusability to a whole new level, this method involves spraying seawater into the atmosphere to increase cloud reflectivity. The extra condensation nuclei created by the spray will change the size distribution of the drops in existing clouds, essentially making them thicker and whiter.

Space mirrors, lenses and moondust

Quite obviously, space based solutions for solar radiation management are deemed unfeasible in comparison to the other SRM techniques. But due to their effectiveness and ingenuity, they're definitely worth the attention:

- **Space mirrors:** having one large or several small (mesh) mirrors floating around the planet to act as reflectors to a certain percentage of sunlight without causing complete darkness underneath.
- **Space lens:** using dispersive methods in space has been proposed



A representation of aerosol injection
using balloons

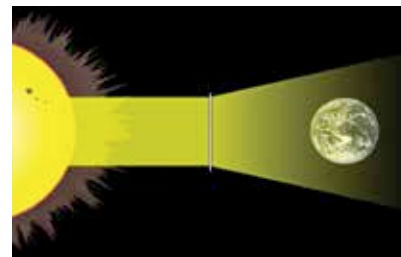
quite a while back. Back in 2004, physicist and sci-fi writer Gregory Benford calculated that a concave rotating Fresnel lens a few mm thick but 1000 kilometres across located at the L1 point between the earth and the sun would reduce the solar energy reaching the Earth by



Difference between the reflectivity of a normal cloud and a 'brightened' cloud

ROADBLOCKS

As with any major scientific project, most climate engineering plans also face their own share of roadblocks even before they've properly begun. To start off, most of these require coordination and communication between multiple nations across the planet – and we've seen how pathetic we are at that! Some of these, like injection of sulphates in the upper atmosphere,



Representation of a space lens and its effect

could be highly risky for particular nations depending on where it is done (above the equator, near the poles, etc.).

Additionally, most of these projects have significant risks and side effects. For example, the same sulphate injection plan could cause even worse warming if it were to be stopped, due to any possible reason. Which lays further emphasis on how important precise coordination around the world is for this. But sadly, in a world where different nations, developing and developed, superpowers and minnows, cannot reach an agreement about the same measurement units of climate change, there really is no hope to a collaboration towards saving this little blue dot that we call home! 🇺🇸